LegalEase

AI-Powered Legal Document Generator

Generative AI with Google Gemini

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Complete Project Documentation & Implementation Guide

Frontend • Flask Backend • Google Gemini AI • SQLite • PDF Generation

1. Project Abstract

LegalEase is a web-based Generative AI application that helps users prepare structured drafts of common legal documents. The application uses Google Gemini to transform user-provided information into a readable document draft. The generated document can be reviewed, edited, stored in a database and downloaded as a PDF. The project demonstrates the practical use of Generative AI with Google technologies in a useful document-automation workflow.

2. Project Title

LegalEase: AI-Powered Legal Document Generator

3. Problem Statement

Creating legal documents manually can be repetitive and difficult for users who are unfamiliar with document structure. LegalEase provides a guided interface where users select a document type and enter relevant information. Generative AI then assists in organizing the information into a structured draft.

4. Objectives

The objectives are to build a simple legal-document generator, integrate Google Gemini for Generative AI, collect information through structured forms, generate readable drafts, save documents for later access, and provide PDF download functionality.

5. Key Features

• Document-type selection
• User information form
• Google Gemini AI generation
• Generated-document preview
• Document history
• SQLite database
• PDF download
• Responsive web interface
• Fallback template when Gemini is unavailable
• Legal/general-information disclaimer

6. Technology Stack

Frontend: HTML5, CSS3, JavaScript
Backend: Python Flask
Generative AI: Google Gemini API using the Google GenAI Python SDK
Database: SQLite with Flask-SQLAlchemy
PDF: ReportLab
Configuration: python-dotenv

7. System Architecture

The system follows a simple client-server architecture:

User Interface → Flask API → Google Gemini → Generated Draft → SQLite Database → PDF Generator

The browser collects document details and sends them to the Flask backend. The backend validates the request, calls Google Gemini when an API key is configured, saves the generated result and returns it to the frontend. ReportLab is used to create the downloadable PDF.

8. Main Modules

Module	Description
Document Selection	Choose Rental Agreement, Affidavit, NDA, Employment Agreement, Complaint Letter or Leave Letter
User Input	Collect name, other party, address, date, purpose and terms.
Gemini AI	Generates a structured draft from supplied information.
Preview	Displays the generated document for review/editing.
Database	Stores document type, input data, generated content and creation time.
PDF Generator	Creates an A4 PDF from the generated draft.
History	Displays recently generated documents and opens saved drafts.

9. Google Gemini Integration

LegalEase uses Google's Gemini Generative AI through the official Python GenAI SDK. The backend sends a carefully structured prompt containing the selected document type and user-supplied information. The prompt instructs the model not to invent missing facts and to use placeholders where important information has not been supplied.

Example Gemini integration:

```
from google import genai
client = genai.Client(api_key=GEMINI_API_KEY)
response = client.models.generate_content(
    model=GEMINI_MODEL,
    contents=prompt
)
document = response.text
```

10. Database Design

Field	Type	Purpose
id	Integer	Primary key
document_type	String	Selected document category
title	String	Document title
input_data	Text	Stored form information as JSON
content	Text	Generated document
created_at	DateTime	Creation timestamp

11. Application Workflow

1. Open the LegalEase website.
2. Select a legal-document type.
3. Enter the available user and document information.
4. Click Generate with Gemini.
5. Flask sends a controlled prompt to Google Gemini.
6. Gemini returns a structured draft.
7. The draft is saved to SQLite.
8. User reviews or edits the generated content.
9. User downloads the document as an A4 PDF.

12. Project Folder Structure

```
LegalEase_Gemini_Project/
■■■ app.py
■■■ requirements.txt
■■■ .env.example
■■■ README.md
■■■ templates/
■ ■■■ index.html
■■■ static/
    ■■■ style.css
```

13. Installation

Install Python 3.11 or newer.

Create a virtual environment: python -m venv venv

Activate it: venv\Scripts\activate (Windows).

Install packages: pip install -r requirements.txt

Copy .env.example to .env.

Add your Google Gemini API key to GEMINI_API_KEY.

Run the application with: python app.py

Open http://127.0.0.1:5000 in a browser.

14. API Endpoints

GET / — loads the LegalEase interface.

POST /api/generate — generates and stores a draft.

GET /api/documents — retrieves recent document history.

GET /api/documents/<id> — opens a saved document.

POST /api/documents/<id>/pdf — creates a downloadable PDF.

15. Testing

Test	Expected Result
Application launch	Home page loads
Document selection	Selected type is accepted
AI generation	Gemini-generated draft appears
Database storage	Document appears in history
History open	Saved content is displayed
PDF download	A4 PDF is downloaded
No API key	Fallback draft is generated

16. Future Enhancements

- User registration and secure login.
- Admin dashboard.
- MySQL or PostgreSQL for multi-user deployment.
- Tamil and English interface.
- More document templates.
- Document version history.
- Cloud deployment.
- Professional legal-review workflow.
- Better validation and security controls.

17. Advantages

- Reduces repetitive drafting work.
- Provides a guided user experience.
- Demonstrates practical Generative AI usage.
- Produces downloadable PDF output.
- Can be extended to many document categories.
- Suitable for an academic Generative AI project demonstration.

18. Conclusion

LegalEase demonstrates how Google Gemini can be integrated into a full-stack application to automate document drafting. The combination of a Flask backend, structured web forms, Generative AI, SQLite storage and PDF generation provides a complete academic project workflow. The architecture is intentionally simple so that it can be demonstrated, understood and extended by students.

19. Legal Disclaimer

LegalEase generates AI-assisted drafts for educational and general-information purposes. AI-generated text can contain errors and may not reflect the law applicable to a particular location or situation. Users should verify all details and seek qualified legal advice before relying on a document for an actual legal matter.

20. Project Demo Sequence

Home Page → Select Document → Enter Details → Generate with Google Gemini → Preview → Save → Download PDF